

The Effect of Short Videos on the Education Outcomes and the Ideology of County-Level Secondary School Students: Evidence from Huitong, China*

Linxi Zeng[†]

The Chinese University of Hong Kong, Shenzhen

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Abstract

This paper conducts a survey regarding phone use, consumption of short videos on phones, and teenagers' ideologies on a sample of 2261 secondary school students from Huitong County in Hunan, China. Results show that consuming short videos comprise a major activity on phones for these students. We test how various factors such as preferred short video platforms, time spent on watching, and preferred content categories contents influence students' opinion polarization level and aspects of students' ideology including political ideology, social ideology, and life ideology. We identify the online friend network beneath short video platforms as a potential subject deserving of scholarly attention.

Keywords: Social Media, Short Video Platforms, Addiction, Chinese County Education

*This paper is submitted to ECO4280 Thesis in Economics II. This is an unfinished version as data are still incoming by the course paper submission deadline, among which is the educational outcome (i.e., test score) data. Therefore only the effect on ideologies is discussed in this version. I plan to work on the current project long-term beyond graduation.

[†]Linxi Zeng: The Chinese University of Hong Kong, Shenzhen, email: linxizeng@link.cuhk.edu.cn. Advisor: Yue Cao.

I Introduction

In 2023, TikTok had 1.5 billion monthly active users worldwide and over 780 million daily users in China Iqbal (2024). Compared to traditional social media platforms, short video platforms like TikTok and Kuaishou have been remarkably successful in reaching and retaining young audience at a time period in their lives where ideologies are nascent in formation. This paper explores the effect of short videos on teenagers' ideologies, using a first-hand survey run on over 2,000 students in a secondary school in Huitong, China. It contributes to research on a highly popular medium that has received little empirical attention (Wang, 2020).

There are very few existing literature in economics studying the new social media form of short videos. But the intersection of social media addiction, mental health, and educational outcomes is well-researched (Allcott, Gentzkow and Song, 2022; DellaVigna and La Ferrara, 2015; Gentzkow, Shapiro and Stone, 2014; Acemoglu, Ozdaglar and Siderius, 2021; Braghieri, Levy and Makarin, 2022; ?). In other social science fields, relatively more literature has focused on short video platforms. Radin and Light (2022) and Escamilla-Fajardo, Alguacil and López-Carril (2021) study the potential pedagogical benefits of incorporating TikTok into classroom teaching. Rand and Brushett (2023) presents TikTok as a casual method to alert students of course updates during COVID-19. Vaterlaus and Winter (2021) qualitatively explores young adults' motivations, experiences, and perceptions of TikTok in relation to Uses and Gratification theory. Through bibliometric and topic modelling analyses, Rejeb et al. (2023) observe that health journals dominate in TikTok research and identify five major clusters. Keywords for the first cluster are TikTok, COVID-19, and public health. The second cluster is around alternative social media platforms, health communication, and education. The third cluster zooms in on TikTok and China, related to algorithms, platform governance, geopolitics, and identity. The fourth cluster studies aspects of social networks, including body image, uses and gratifications (U and G), motivation, personality traits, and self-esteem. Finally, the last cluster emphasizes the popularity of TikTok among young people. Given that these groups remain central, they could be vulnerable to predators ranging from radicals, terrorists, and cyber-attackers, to criminals.

This paper is going to draw on literature from both inside and outside of the economic academic journals, and test some well-known hypotheses as well as suggest new ones.

II Data

Huitong is a county located near the west border of Hunan province. In 2023, its reported Gross Regional Product (GRP) is 10.6 trillion yuan, the number of its registered population is 364 thousand, and the number of its permanent residents is 283 thousand (?). It was classified as an "impoverished county"¹ until 2018 Hunan Bureau of Education (2021). The secondary level education in Huitong comprises of two normal secondary schools: Huitong County No. 1 Secondary School, Huitong County No. 3 Secondary School, and a technical school,

¹The classification criteria include per capita county GRP, per capita county general budget revenue, per capita net income of county farmers, and other indicators highly correlated with poverty levels. Special considerations are given to old revolutionary base areas, ethnic minority areas, and border areas (Weekly, 2020).

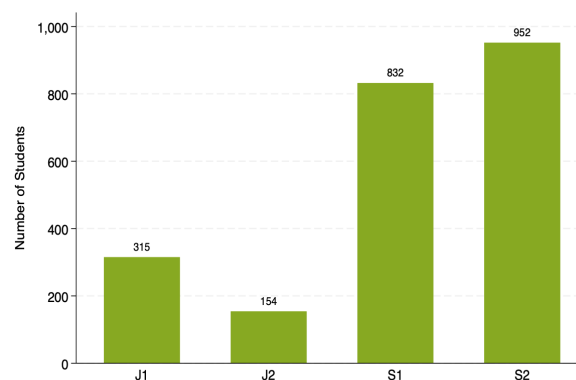
Huitong County Vocational Technical School. No. 1 outruns No. 3 in Gaokao outcome by a wide margin, and it is the target survey school in this paper (henceforth, "the school"). The school's educational success has received provincial media acclamation (Duan, Ming and Gao, 2018), and it has maintained a record of sending one to two students to the top two universities in China, Peking University and Tsinghua University, every two years.

The school has both a middle and a high school division, each including three grades. I collect information on students' usage of cellphones in general and of short-video platforms in particular, their attitudes toward this usage, and various aspects of their ideological beliefs.

The survey is designed on the online platform Wen Juan Xing², and it is distributed to each student via a school computer at the beginning of computer classes, administered by teachers. This ensures a near 100% response rate but excludes the third-year students in both middle and high schools from the sample, since computer classes are not offered in these grades to facilitate full concentration on the preparation for the Zhongkao and the Gaokao.

The first round of surveys are collected in the span of three weeks (April 8 to April 25, 2024). The sample size is 2261. It is roughly balanced across genders with females comprising 51% of the population, males 48%, and non-binary 1%. Of the 21 students who identify themselves as non-binary, most of them specify their gender to be "Walmart shopping bag," "gunship," "robot," or "ladyboy," suggesting that they could be appealing to viral memes instead of seriously stating their gender identity. Therefore in some of the upcoming analyses I drop the "other" gender category. The distribution across grades is imbalanced due to multiple reasons, as is shown in Figure 1. The average grade size in middle school is around 400, and that in high school is around 1050. As of April 29, some classes are still absent from the sample because their computer classes have been suspended, which are in turn largely due to computer teachers being assigned alternative exam invigilator duties.

Figure 1: The Distribution of Student Sample Size Across Grades



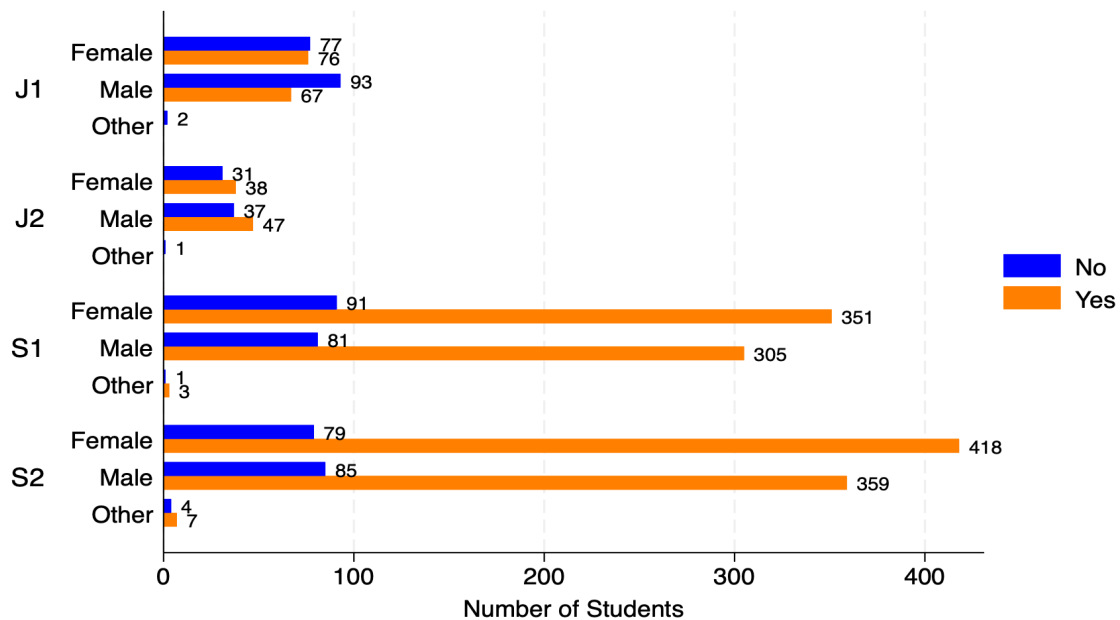
Notes: This figure plots the number of surveys collected from each of the four grades in the school, with the x-axis arranged from lower to upper grades (J1 and J2 are middle school 1st and 2nd grades, where J stands for junior; S1 and S2 are high school 1st and 2nd grades, where S stands for senior). Third-year students in both middle and high schools are not surveyed.

Figure 2 presents the number of students who reportedly have their own smartphones. Smartphone ownership among students increases sharply from middle to high school. For both grades in middle school, around half of

²<https://www.wjx.cn/>

the students have their own smartphone. In high school, this rate rises to about 79% for first-years, and 83% for second years. The percentage of students who have their own smartphones is slightly higher for females than for males.

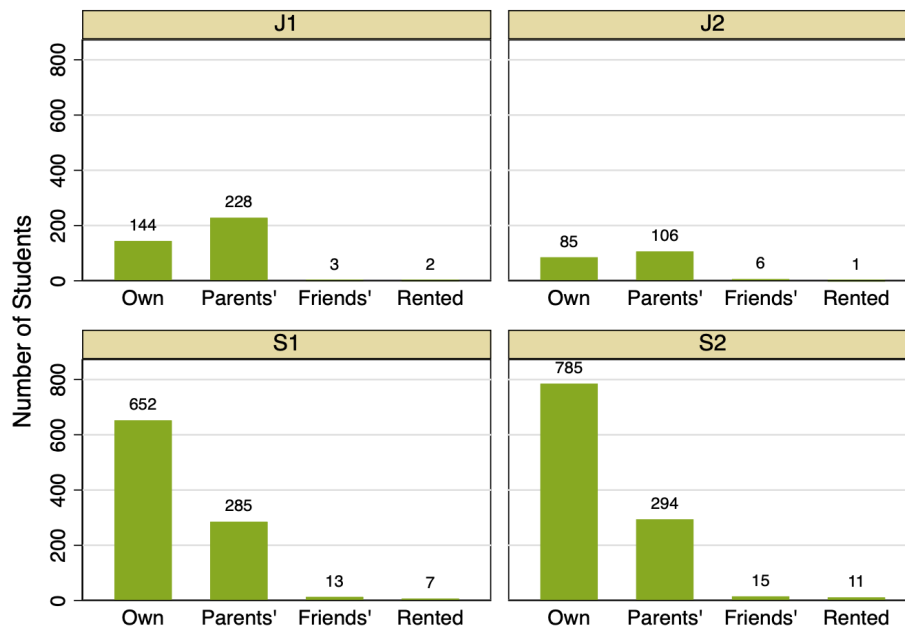
Figure 2: The Number of Students Who Have Their Own Smartphones by Gender and Grade



Notes: This figure plots the number of students who report to have their own smartphones by gender and grade.

Figure 3 shows that middle school students use parents' phones more than their own, which is very likely due to not yet owning a smartphone (refer to Figure 2). 1/5 to 1/4 of high school students still use parents' phones but there is possibly a larger "clean" transition from only using parents' phones to only using their own phones than a "sticky" transition where they maintain using parents' phones even when they come to have their own phones. The differentiation between clean and sticky transition is interesting because using parents' phones allows for a larger intergenerational influence on content consumed on TikTok, WeChat feeds and subscriptions, etc., if these channels are present. A clean transition is more understandable than a sticky transition for multiple reasons, convenience, fewer restrictions, teenagers' growing sense of privacy and individuality, etc. Figure 3 also disproves a hypothesized widespread phenomenon of students renting phones from retail stores near school, as stated by an interviewee who is an English teacher in Huitong County No. 3 Secondary School. It is possible that some students conceal this behavior in the survey, but the visible number of students who use rented phones is extremely small.

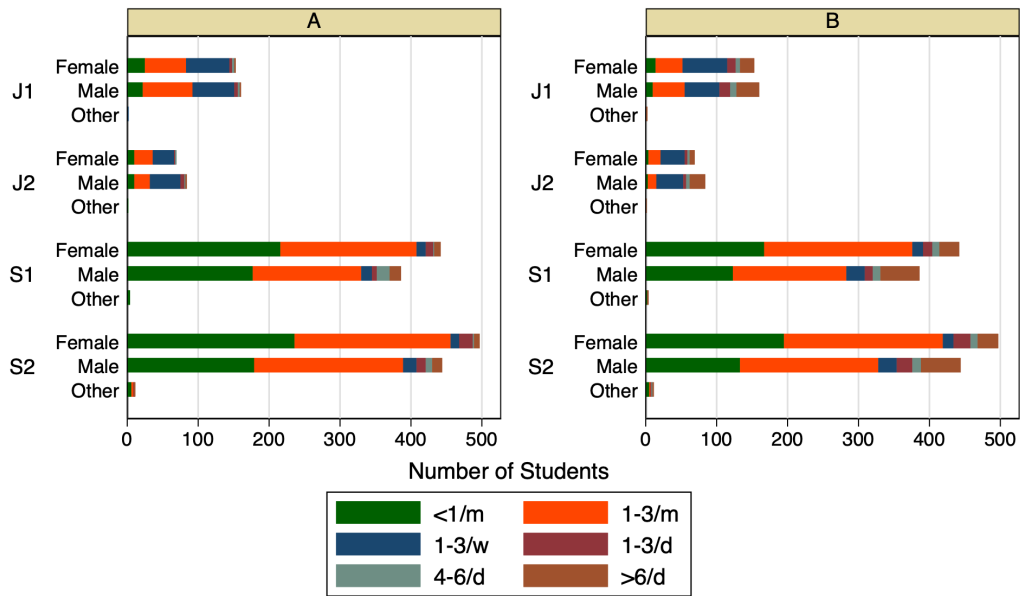
Figure 3: Types of Phones Used by Students



Notes: This figure plots the types of phones students use by grade.

On the high school level, more than 80 percent of students report to use phones fewer than three times per month. There is a slight increase in frequency from the first year to the second year students. And females report an overall lower frequency of phone usage than males. The middle school sample is likely too small to provide reliable evidence, but only around half of the students use phones fewer than three times a month, and the other half, more than once a week. One may hypothesize that middle school students use phones more frequently than high school students. But I do not think this is a reasonable hypothesis, and a more likely explanation for the increase in percentage of students who use phones more than once per week is that teenagers who attend the high school is often a superset of those who attend the middle school, and the middle school students are on average financially better off than the high school students. A final note is that regardless of grade and gender, students seem to overestimate the phone use frequency of their classmates compared to self-reports, as shown by the overall leftward shift of color bars from panel A to panel B.

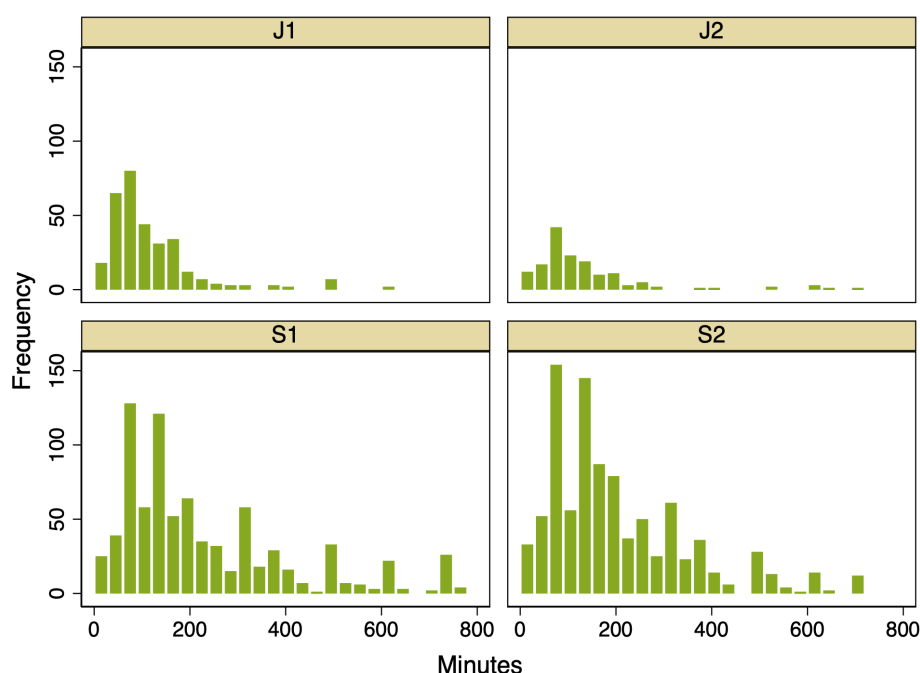
Figure 4: Phone Use Frequency



Notes: This figure plots the reported phone use frequency by grade and gender, where panel A records self-reports and panel B records perceived frequency of classmates. Month, week, and day, are abbreviated to m, w, and d.

We can see from Figure 5 that the duration of each phone use clusters around one to three hours, with two peaks at one and two hours. The interesting slump in between is likely due to how the survey is designed, where students are asked to fill in the fields with units in hours and minutes separately.

Figure 5: Phone Use Intensity (Duration Each Time)



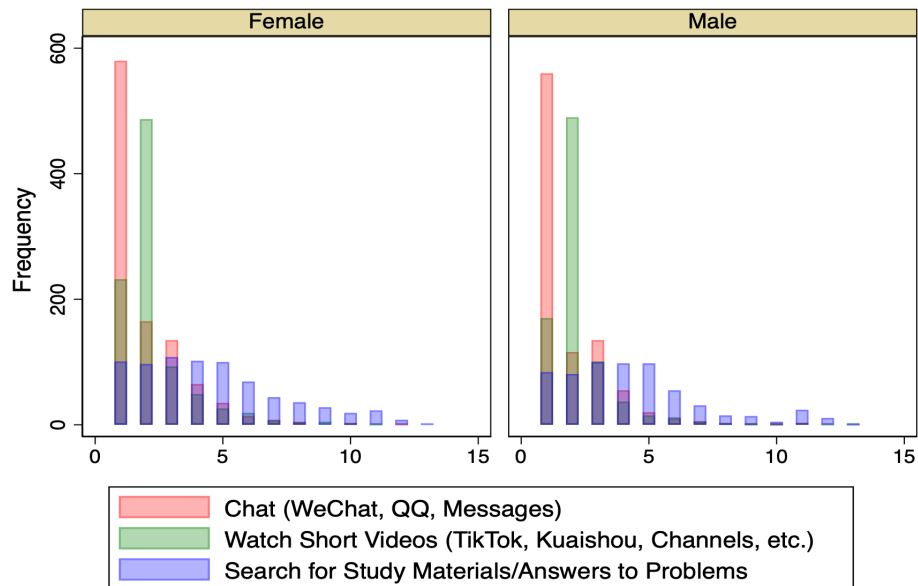
Notes: This figure plots the duration time of each phone use by grade. Answers with the number of hours greater than 12 are excluded from the graph. The width of each bin is 30 minutes.

From Figure 6 we see that almost 60 percent of all students rank chatting as their most-used phone function. While 55 percent of all students rank watching short videos on platforms such as TikTok (i.e., "Douyin" in China), Kuaishou, Channels (affiliated to WeChat), etc., as their second most-used function, around 20 percent rank it as their most-used function. This testifies that short video platforms have successfully infiltrated the school's students' lives, students spend a significant amount of phone time on consuming short videos, and it reiterates the social significance of research on short videos and their impact on teenagers. The distributions of the top three functions do not exhibit obvious differences between genders. However, gender differences emerge in Panel (b) of Figure 6 among functions that comprehensively rank between four and six. There are almost 100 female students who rank reading novels as their first, second, ..., and fifth choices, each; whereas there are around 200 male students who rank playing games as their first, second, and third choice, each. This provides suggestive evidence for a teacher interviewee's observation that boys tend to be addicted to games and girls to novels, but it would be false to say these functions dominate the consumption of short videos. Browsing social media is mostly ranked as the third most-used function. This is surprising because this lends evidence to the speculation that short video platforms have surpassed traditional social media platforms by a wide margin in popularity among teenagers. Taking photos or videos mostly ranking the fourth among both girls and boys is also interesting. The surroundings for these students are highly monotonous from day to day, it is reasonable to guess they mostly take pictures or videos of themselves or other people, which they might post online or send to others. But this warrants more investigation in this alley. In comparison, reading news, shopping on Taobao, and watching livestreams play

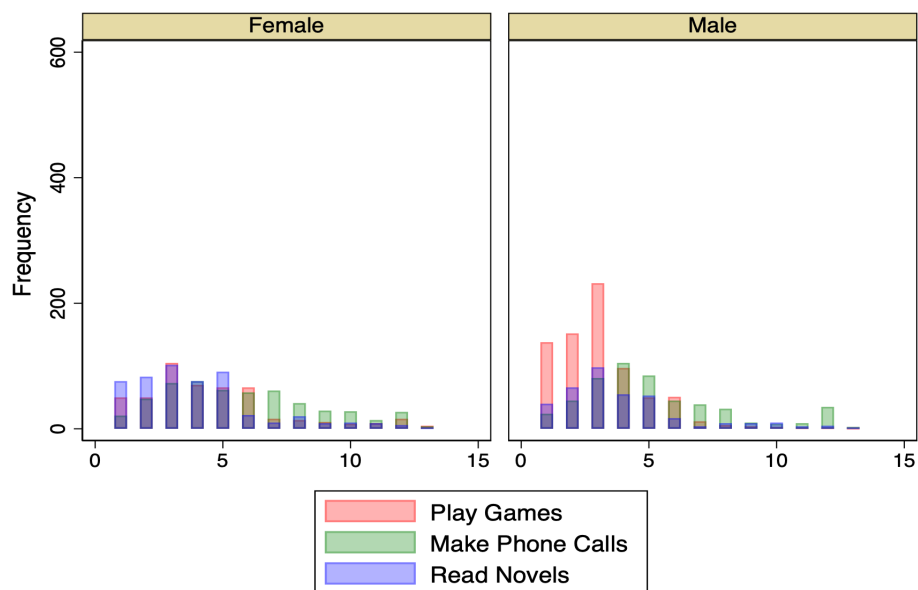
a marginal role in students' phone use.

Figure 6: Usage Frequency Ranking of Phone Functions

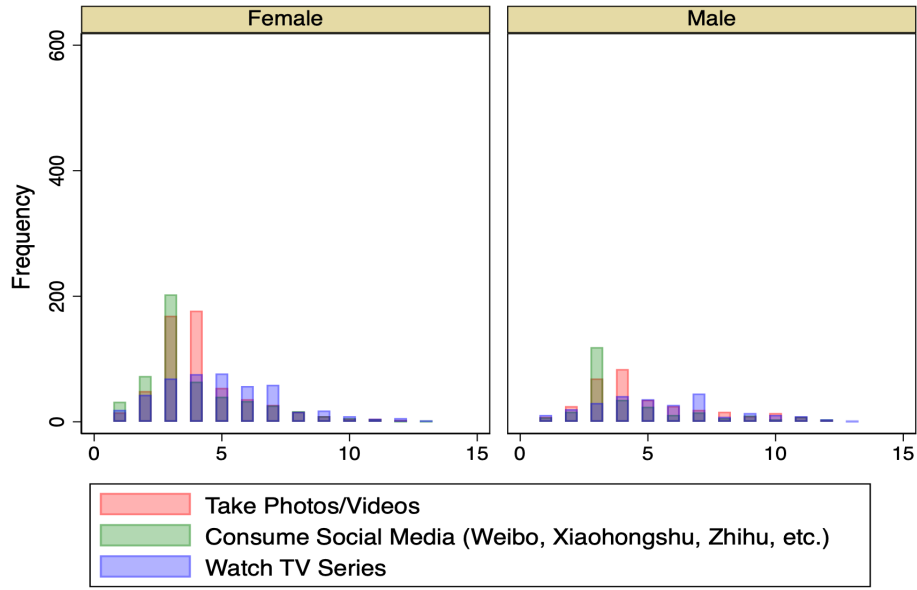
(a) No. 1-3



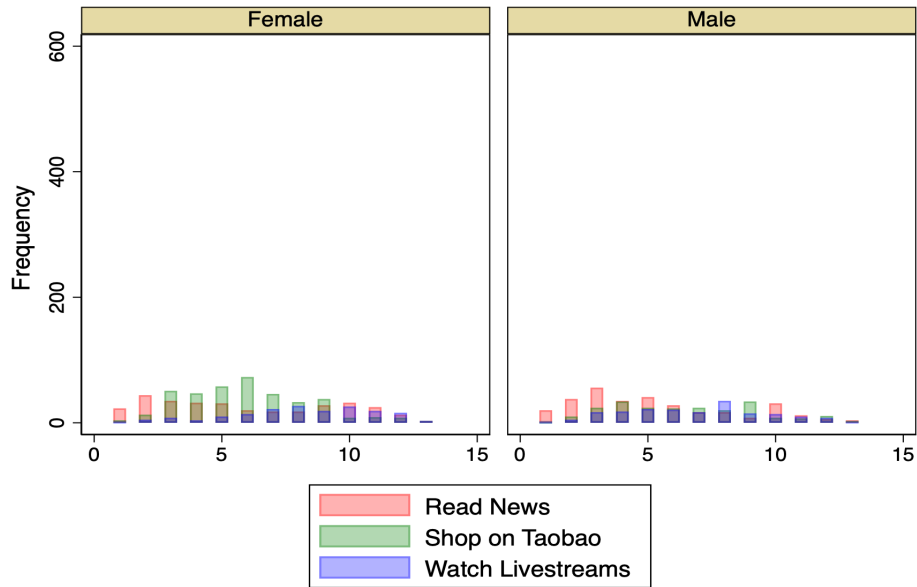
(b) No. 4-6



(c) No. 7-9



(d) No. 10-12

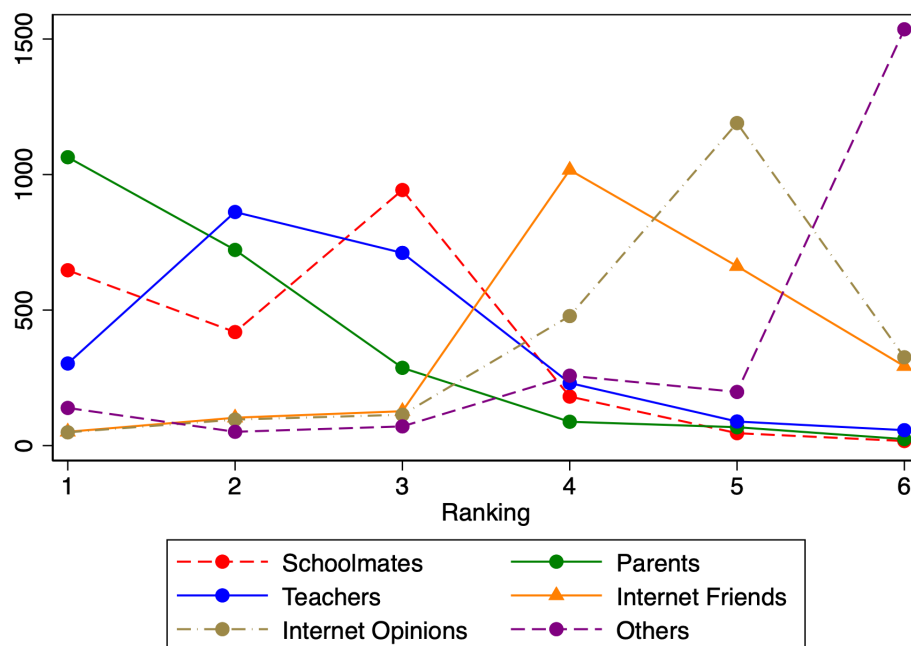


Notes: These figures plot the usage frequency ranking of different phone functions. The survey question asks students to rank among 13 functions according to usage frequency, choosing at least three functions to rank. The x-axis denotes the ranking. The ordering of functions as appeared in the subfigures is determined by a comprehensive score of each function, calculated by $(\sum \text{Frequency} * \text{Weights}) / \text{Number of Students Selecting The Function}$, where *Frequency* are the frequency of a given function in each ranking (1st, 2nd, 3rd, etc.) and *Weights* are in an inverse order of the rank of a given function (e.g., in this case, the weight of the 1st place is 13, 2nd, 12, 3rd, 11, ..., and the weight of the 13th place is 1.). The 13th choice "others" and is excluded from the graphs since it has the lowest comprehensive score and is unspecified. The gender "Other" is also excluded for clean presentation.

Overall, Figure 7 shows that students report to give the following sources of opinions descending levels of trust: parents, teachers, schoolmates, internet friends, other internet opinions, and others. It is interesting that their trust in internet friends rank just below that for schoolmates, suggesting internet friends to be a potential source of ideology influence. However, this survey question requires that students include all given choices in the

ranking, therefore it provides limited evidence for how much internet friends influence teenagers. On the other hand, another survey question directly asks whether students make online friends through short video platforms, and as Figure 8 shows, almost 3/4 of both males and females in all grades answer in the affirmative. The online friends network is hidden behind short video platforms and is worth academic attention. Short video platforms enable people to connect with short video producers and other followers (especially if the producer also engages in livestreaming), and sentiments like admiration for an internet friend's skill in a specific field (e.g., gaming skills) may induce higher susceptibility to influence from this internet friend in other aspects of one's ideology. For example, it is increasingly common for actors or gamers to influence their fans' political ideology by being politically vocal. It is important to acknowledge and study the phenomenon of establishing a personal "IP (Intellectual Property)" or brand through excelling in one field, gaining attention, and then spilling power over and exerting influence in other fields, which is made easier by the age of self-media that the form of short videos helps to solidify.

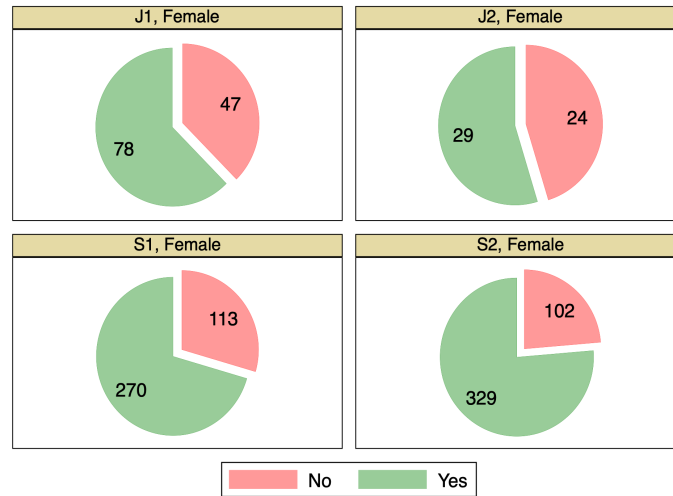
Figure 7: Ranking of Trust Level in Different Sources of Opinion



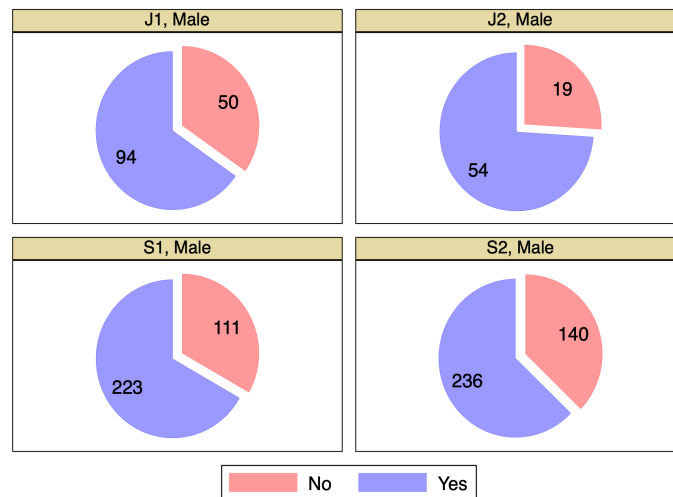
Notes: The corresponding survey question asks students to rank their trust in different sources of opinion from high to low, and this figure plots the number of students in each (ranking, opinion source) pair.

Figure 8: Whether Students Make Online Friends Through Short Video Platforms

(a) Females



(b) Males

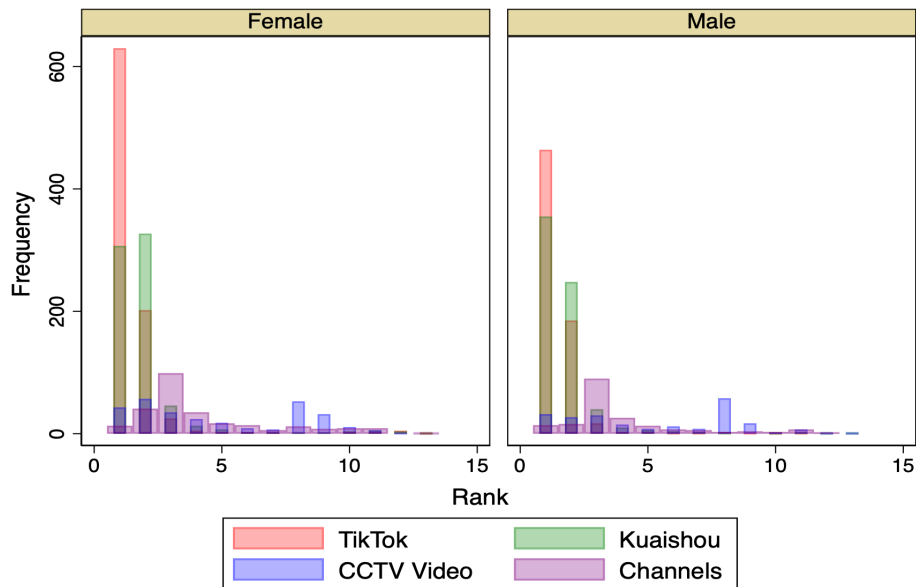


Notes: These pie charts plot the proportion and number of students who make or do not make online friends through short video platforms by gender and grade. The gender "Other" is excluded for clean presentation.

Calculating comprehensive scores of each choice in the same way as written in the notes of Figure 6, Figure 9 plots the distribution of the number of students putting each of the top four choices in each ranking (1 to 13). The first and second most-used platforms are the wellknown TikTok and Kuaishou. In the third and fourth places are CCTV Video and Channels, operated by CCTV Media Development Co., Ltd., and Tencent's WeChat, respectively. I find that of all the 2253 students who effectively put down any first choice, almost 50 percent put down TikTok as their most-used short video platform, 30 percent put down Kuaishou, eight percent choose "others," and seven percent choose "I have never watched a short video." A larger proportion of males put Kuaishou as their first choice than females. Of the 1330 students who effectively put down any second choice, 30 percent state TikTok, 45 percent state Kuaishou, and the rest of the sample are roughly evenly spread out among the 11

other choices. Starting from the third choice onward, the number of students who indicate a choice in that rank declines from 600 to 100. There is a stable 120 people who assign each ranking to the choice with an identical order as that ranking when listed in the survey, and the rest of the sample spreads out around that choice in bell-shape. Therefore I suspect that these students mechanically ranked the choices in the given order, which does not reflect their true preference.

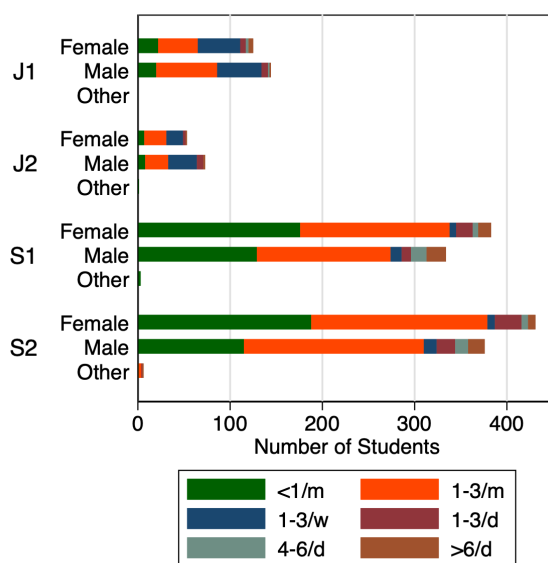
Figure 9: Ranking of Short Video Platforms According to Usage Frequency



Notes: The corresponding survey question asks students to rank 11 short video platforms and choices "others" and "I have never watched a short video" according to usage frequency, and students are asked to select at least one choice to rank from the list. This figure plots the number of students in each ranking of the four choices with the highest comprehensive scores (the formula for the computation of comprehensive scores is described in the notes of Figure 6).

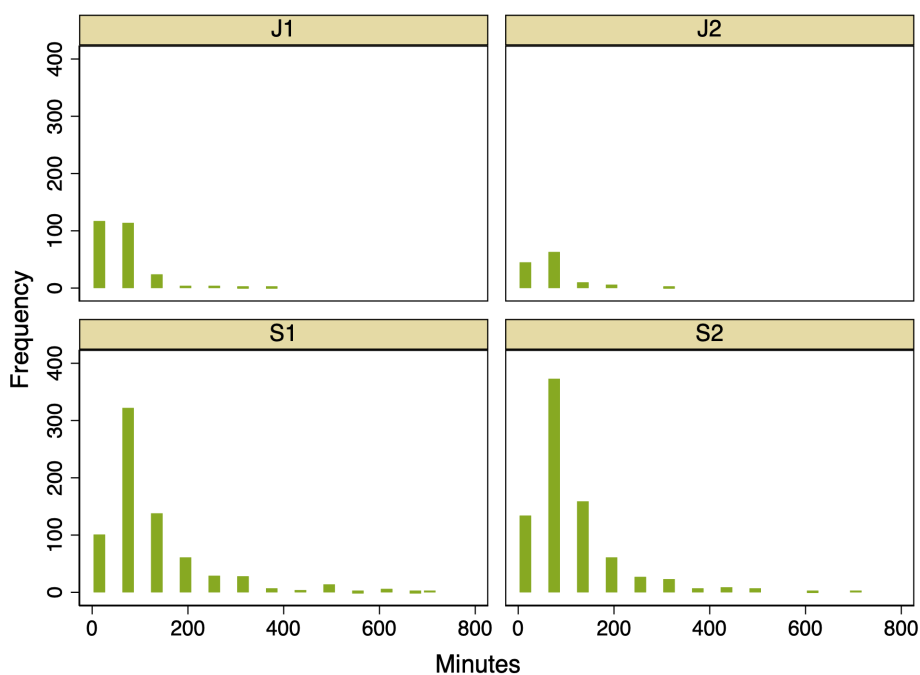
From Figures 10 and 11 we see the vast majority of high school students view short videos fewer than three times per month, and the duration of each viewing session centers around one to 1.5 hours, with a certain amount of students in the "less than one hour" and "between two to four hours" bins. Most middle school students indicate their viewing intensity to be less than 1.5 hours each time, but almost half of them view short videos more than once per week, shifting the overall distribution of viewing frequency to the right side of that for high school students. However, we cannot conclude that middle school students view short videos more frequently than high school students without additional information.

Figure 10: Short Video View Frequency



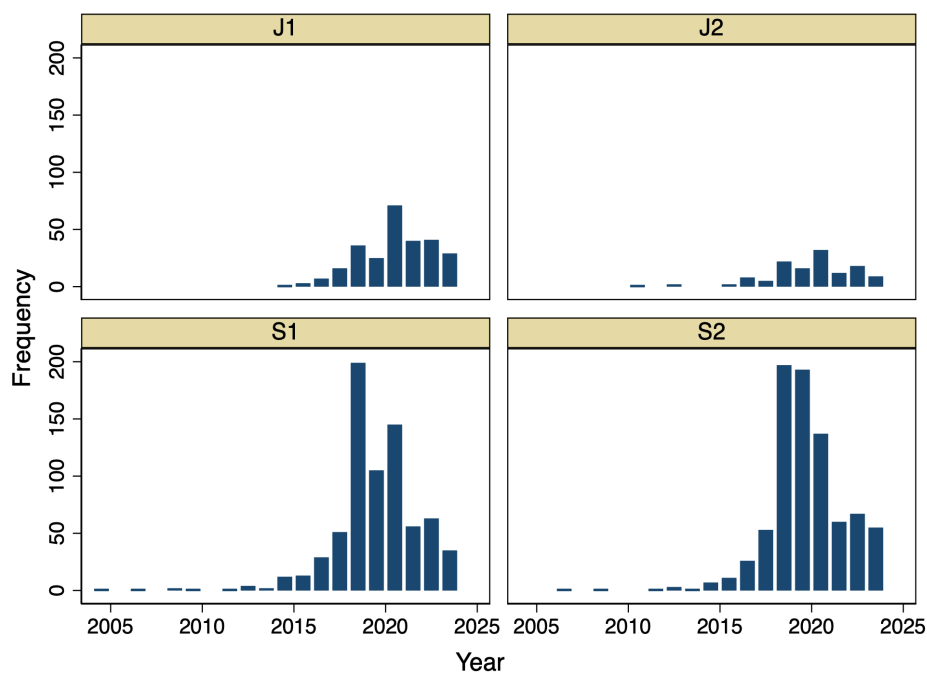
Notes: This figure plots the reported short video viewing frequency by grade and gender. Month, week, and day, are abbreviated to m, w, and d.

Figure 11: Short Video View Intensity (Duration Each Time)



2011 (Cheng, 2021) and Douyin in 2016 (Yeung and Wang, 2023). This survey question was inspired by previous studies employing staggered rollout of TVs (Gentzkow and Shapiro, 2008) and Facebook (Braghieri, Levy and Makarin, 2022) to identify causal effects on student educational outcomes and mental health. The introduction of short video platforms does not seem to follow a clean rollout process, but there may still be staggered timing in when short video platforms populate a certain local market. While it is true that short videos can go viral across the internet very quickly and that the availability of short video apps does not discriminate between different domestic locations, it is also unlikely that all cities and towns welcomed these apps simultaneously. There could still be some timing heterogeneity here that could be exploited for research purposes, but this heterogeneity was either intentional (like how Braghieri, Levy and Makarin (2022) exploits the staggered Facebook rollout across US colleges between February 2004 and September 2006 before being made fully public) but which decisions I am yet unaware of, or it was unintentional, in which case perhaps the only way to solicit this timing data is through surveys in local markets.

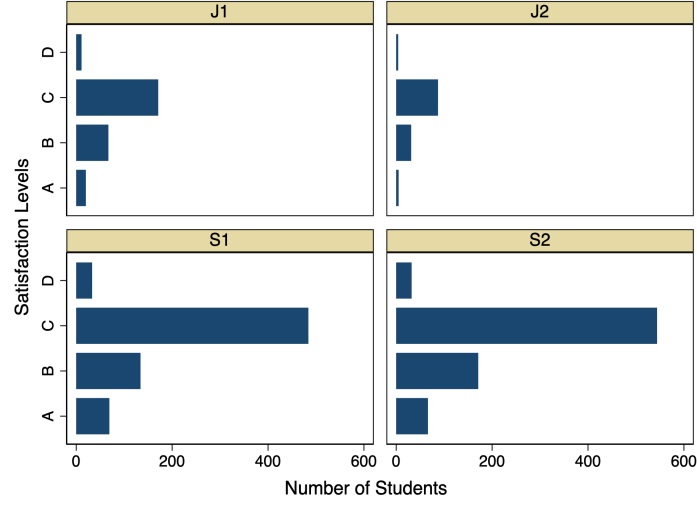
Figure 12: Year Introduced to Short Video Platforms



Notes: This figure plots the year in which students were introduced to short videos by grade. Students are asked to provide their best estimate if they cannot recall with certainty.

Regarding self-evaluated sentiments toward one's own short video consumption, most of the surveyed students report to be satisfied with their current situation (option C in Figure 13). About two percent of students in each grade thinks they would be happier if they could spend less time on short videos (option B in Figure 13), indicating potential self-detected short video addiction issues. Less than one percent of student wish they could spend more time on short videos. And the students who choose "other" (option D in Figure 13) mainly specify an attitude that is more neutral or which involves more critical thinking.

Figure 13: Students' Satisfaction Level Toward Current Short Video Consumption



Notes: This figure plots the self-evaluated satisfaction level toward current short video consumption by grade. A: Not satisfied, I would be happier if I could spend more time on watching short videos. B: Not satisfied, I would be happier if I could spend less time on watching short videos. C: Satisfied, I am very happy with my current situation. D: Other.

I then attempt to register various aspects of students' ideology through making a series of opinion statements and asking students to rate to what extent they agree with the statements. They are given five choices for each statement, "strongly disagree," "relatively disagree," "neutral," "relatively agree," and "strongly agree." The downside to asking students directly for their attitudes and beliefs is that there is no way to ensure truthful reporting, which comprises one of the biggest challenges and limitations of this paper. For most of these questions, between 40 and 60 percent of students check "neutral," indicating potential moderacy response bias (Stantcheva, 2022), but we cannot discount the possibility that these students might in fact be neutral toward these statements.

In the following sections I will specify a simple regression model and provide results of statistically tests regarding short video and teenage ideology.

III Empirical Strategy

Due to data constraints, we employ a simple linear regression with a model specified as follows:

$$Y = \alpha + \beta Z + \gamma X + \varepsilon$$

where Z is our independent variable of interest, X is a vector of controls (i.e., gender and grade), and ε is the error term. The two kinds of outcome variables this paper will discuss are 1) polarization indicator, and 2) ideology.

IV Results

A Opinion Polarization and Overall Short Video Consumption

In this section we explore the effect of different factors on the extent to which an individual's opinion is polarized or tends to be extreme. Since only five choices from "strongly disagree" to "strongly agree" are offered for the 42 opinion questions in the survey, then denoting them using integers 1 to 5 and drawing from (Bramson et al., 2017), we can define the level of opinion polarization in at least the following ways: 1) spread, where we can use the range of each student's choices as a proxy for polarization level; and 2) dispersion, where we can use the standard deviation of each individual's answers across 42 questions as a proxy for polarization level.

Regression results on indicators of overall short video consumption are shown in Tables 1 and 2. We see that short video consumption frequency and the number of years since first exposure to short videos are positively correlated with both opinion range and opinion standard deviation, signaling a higher degree of polarization.

Table 1: Opinion Range And Overall Short Video Consumption

	(1) Opinion Range	(2) Opinion Range	(3) Opinion Range	(4) Opinion Range
Frequency	0.0641*** (0.0227)			
Intensity		0.000271* (0.000142)		
Freq. * Int.			0.0000751 (0.0000596)	
Years				0.0670*** (0.0161)
Gender	-0.274*** (0.0540)	-0.262*** (0.0541)	-0.264*** (0.0541)	-0.288*** (0.0538)
Grade	-0.148*** (0.0171)	-0.160*** (0.0172)	-0.158*** (0.0171)	-0.170*** (0.0175)
Observations	1919	1919	1919	1919
R^2	0.046	0.044	0.043	0.057
Adjusted R^2	0.045	0.042	0.042	0.056
rmse	1.172	1.174	1.174	1.166

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table presents the regression results of opinion range on short video view frequency, intensity (duration each time), their product, and the number of years since first introduced to short videos, controlling for gender (excluding "other") and grade.

However, all these variables are only rough indicators of what they are presented to represent. More controls are also needed to yield more convincing evidence on how overall short video usage affects opinion polarization.

Table 2: Opinion Standard Deviation And Overall Short Video Consumption

	(1)	(2)	(3)	(4)
	Opinion SD	Opinion SD	Opinion SD	Opinion SD
Frequency	0.0207*** (0.00745)			
Intensity		0.000104** (0.0000442)		
Freq. * Int.			0.0000316* (0.0000185)	
Years				0.0220*** (0.00479)
Gender	-0.0651*** (0.0167)	-0.0615*** (0.0168)	-0.0624*** (0.0168)	-0.0697*** (0.0166)
Grade	-0.0484*** (0.00567)	-0.0523*** (0.00563)	-0.0516*** (0.00561)	-0.0555*** (0.00570)
Observations	1919	1919	1919	1919
R^2	0.045	0.043	0.043	0.058
Adjusted R^2	0.044	0.042	0.041	0.056
rmse	0.363	0.364	0.364	0.361

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table presents the regression results of opinion standard deviation on short video view frequency, intensity (duration each time), their product, and the number of years since first introduced to short videos, controlling for gender (excluding "other") and grade.

B Preference for Short Video Contents and Ideologies

This section explores how short video contents, i.e., preferences for different short video categories, affect specific ideologies. I first present the list of opinion statements on which students' attitudes are obtained through the survey in Table 3. These statements encompass attitudes toward short video platforms per se, political ideology (nationalism, trust in government, willingness to participate in public discourse, etc.), gender ideology; some statements are more generally about life and opinions, and some are meant to test self-understanding, critical thinking tendencies, and potential echo chamber effects. Statements that are close by nature can be used to cross-test and cross-validate. As ? points out, the political ideologies in China are under-researched and not well-defined, and it does not fall neatly in the left-right western spectrum. Therefore I try to touch on as many aspects as possible in this survey.

Table 3: Index for Opinion Statements in the Survey

Index	Statement
S1	I like the videos recommended to me by short video platforms.
S2	I think watching short videos can increase my understanding of the opposite sex.

Continued on next page

Table 3 – continued from previous page

Index	Statement
S3	I think short video platforms has had more positive impacts than negative impacts on me.
S4	I think watching short videos makes it harder for me to concentrate on studying.
S5	If I went viral overnight, I would be very happy.
S6	I think watching short videos has given me a more independent personality.
S7	... helps me gain more knowledge and better understand the world.
S8	... has increased my negative emotions towards the opposite sex.
S9	... helps me learn what is right and what is wrong.
S10	... brings me more positive emotions than negative emotions.
S11	I like watching short videos more than communicating with real people.
S12	I think watching short videos is a waste of time.
S13	... is an important part of my life.
S14	... helps me explore my sexuality.
S15	... will affect my future career plans.
S16	... makes me believe my own opinions more.
S17	... has increased my acceptance of violence.
S18	... makes me a freer thinker.
S19	... makes me more convinced that there are shortcuts to success.
S20	... makes me question what I learn at school or at home.
S21	... makes me want to hear similar opinions as my own rather than different ones.
S22	... has enriched my social circle and helped meet my social needs.
S23	... makes me more confident that whatever the government does is right.
S24	... makes me care more about my own growth.
S25	... increases my longing for the outside world.
S26	... makes me feel that education in reality is too old-fashioned and traditional.
S27	I also want to be a short video blogger/livestream host.
S28	I think watching short videos increases my desire to buy things.
S29	... makes me hate America even more.
S30	... makes me feel empty and that life is meaningless.
S31	I think opinions in short videos are often more reasonable than what I hear in real life.
S32	I think watching short videos makes me love my motherland even more.
S33	... makes me more willing to publicly express my opinions.
S34	... makes me support relevant policies during COVID-19 even more.

Continued on next page

Table 3 – continued from previous page

Index	Statement
S35	I think there should be more different voices and perspectives in our society.
S36	I think society is fair.
S37	I like to be arranged for rather than having to arrange for myself.
S38	I think I have a lot of passion for life.
S39	I think I am very independent when it comes to opinions.
S40	I feel I have strong national pride.
S41	I think my opinions are mostly right, and opinions different from mine are most likely wrong.
S42	I think rules are man-made and should be broken sometimes.

Theoretically, it would be ideal to test the regression coefficient for each pair of (content category, statement) with suitable controls, i.e., list out $17 * 42 = 714$ regressions. But this is a daunting task for the scope of this paper, so I will choose a few pairs of particular interest and illustrate the regressions.

From Table 4 we can roughly see that a preference for gaming related short videos may increase a teenager’s acceptance toward violence, while a preference for arts or travel related content may make a teenager more “liberal” in the sense that they think there should be more diverse perspectives in society. Due to time constraint, I cannot exploit all potential connections of this kind now. This will be done later.

In addition, we should also study how the preferred content categories may impact the previously defined polarization level, the total time spent on short videos, and educational outcomes. We may also introduce the different usage preferences for short video platforms and study if there are heterogeneous effects of short videos from different platforms.

V Conclusion

This paper conducts a survey regarding phone use, consumption of short videos on phones, and teenagers’ ideologies on a sample of 2261 secondary school students from Huitong County in Hunan, China. Results show that consuming short videos comprise a major activity on phones for these students, supporting the social significance of the current research topic. We test how various factors such as time spent on watching short videos, and preferred content categories contents influence students’ opinion polarization level and aspects of students’ ideology including political ideology, social ideology, and life ideology. We find that the frequency of short video consumption and the number of years since first exposure to short videos are significantly positively correlated with an individual’s opinion polarization. Some preliminary regressions also suggest that a preference for game-related short videos increases one’s acceptance level towards violence. And a preference for arts or travel related content may make a teenager more “liberal” in the sense that they think there should be more diverse perspectives in

Table 4: Selected Opinion Statements and Selected Preferred Short Video Content Categories

	(1) S17	(2) S29	(3) S35
Games	0.0608*** (0.0219)		
Opinions		0.0289 (0.0250)	
Art+			0.0830*** (0.0313)
Food+			0.000165 (0.0223)
Music+			0.0248 (0.0277)
Travel+			0.0697** (0.0275)
Gender	0.162*** (0.0464)	0.0998** (0.0418)	-0.0663* (0.0368)
Grade	0.0470*** (0.0159)	0.0168 (0.0169)	-0.00418 (0.0135)
Observations	1919	1919	1919
R^2	0.020	0.004	0.039
Adjusted R^2	0.018	0.003	0.036
rmse	0.949	0.908	0.777

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table presents the regression results of the extents to which a student agrees with S17: I think watching short videos has increased my acceptance of violence, S29: I think watching short videos makes me hate America even more, S35: I think there should be more different voices and perspectives in our society, on the extent to which a student likes certain categories of short video content (which are listed in the leftmost row), controlling for gender (excluding "other") and grade.

society. However, these are mere example of how the collected survey data could be exploited to discover similar relationships, and this work will be carried on into the future. In addition, we identify the online friend network beneath short video platforms to be a potential subject worthy of future research.

After obtaining educational outcome data, I will be able to add another dimension of variable to this research and test how various aspects of short video consumption (time spent on short videos, preference towards different contents) impact students' test scores. I must also note that the survey recorded in this paper was only meant to be a preliminary survey that assists with the design of more in-depth designs that can better address our research question. Therefore the survey questions have breadth but lack depth. But it was a necessary step to determine some foundational information of this research. For example, it confirmed that enough students are exposed to short videos and that they watch it enough to assure the social relevance of this topic. It would be ideal if this and future rounds of surveys can be rolled out in more secondary schools across China, both rural and urban. The ultimate shortcoming to using surveys to gauge ideologies is that truthful reporting cannot be guaranteed or even

checked, more measures would need to be taken to improve the credibility of the research results. A recent paper, Haaland et al. (2024), may offer some insights into how to address this shortcoming.

This project has not proceeded to the stage of performing rigorous textual analysis on students' subjective text responses yet. But reading their answers, I am astonished by how much critical thinking these children exhibit in their thoughts on short videos, and how overall positive their sentiments towards short videos are. Over and over again, they write, even with gratitude, "short videos show me a world I've never seen before, they make me want to study hard and go out (of the mountains) to see the world outside." Over and over again, I am reminded of what research is truly for, and why I chose this road in the first place. To be humbled by what is seen, to understand what was not understood, and to care for what is real.

References

- Acemoglu, Daron, Asuman E. Ozdaglar, and James Siderius.** 2021. "Misinformation: Strategic Sharing, Homophily, and Endogenous Echo Chambers." *SSRN Electronic Journal*.
- Allcott, Hunt, Matthew Gentzkow, and Lena Song.** 2022. "Digital Addiction." *American Economic Review*, 112(7): 2424–2463.
- Braghieri, Luca, Ro'ee Levy, and Alexey Makarin.** 2022. "Social Media and Mental Health." *American Economic Review*, 112(11): 3660–3693.
- Bramson, Aaron, Patrick Grim, Daniel J. Singer, William J. Berger, Graham Sack, Steven Fisher, Carissa Flocken, and Bennett Holman.** 2017. "Understanding Polarization: Meanings, Measures, and Model Evaluation." *Philosophy of Science*, 84(1): 115–159.
- Cheng, Yu.** 2021. "Kuaishou mops up \$5.32b from float in Hong Kong."
- DellaVigna, Stefano, and Eliana La Ferrara.** 2015. "Economic and Social Impacts of the Media."
- Duan, Xianju, Xing Ming, and Wencheng Gao.** 2018. "A survey of "a county special in education" in Huitong, Hunan." <http://203.192.15.131/content/20180911/Article08002BB.htm>.
- Escamilla-Fajardo, Paloma, Mario Alguacil, and Samuel López-Carril.** 2021. "Incorporating TikTok in higher education: Pedagogical perspectives from a corporal expression sport sciences course." *Journal of Hospitality, Leisure, Sport & Tourism Education*, 28: 100302.
- Gentzkow, Matthew, and Jesse M. Shapiro.** 2008. "Preschool Television Viewing and Adolescent Test Scores: Historical Evidence from the Coleman Study *." *Quarterly Journal of Economics*, 123(1): 279–323.
- Gentzkow, Matthew, Jesse Shapiro, and Daniel Stone.** 2014. "Media Bias in the Marketplace: Theory." National Bureau of Economic Research w19880, Cambridge, MA.
- Haaland, Ingar, Christopher Roth, Stefanie Stantcheva, and Johannes Wohlfart.** 2024. "Measuring What Is Top of Mind."
- Hunan Bureau of Education.** 2021. "Huaihua Huitong County: Give Children Good Education Near Their Homes - Hunan Bureau of Education."
- Iqbal, Mansoor.** 2024. "TikTok Revenue and Usage Statistics (2024)."
- Radin, Abigail G. B., and Caitlin J. Light.** 2022. "TikTok: An Emergent Opportunity for Teaching and Learning Science Communication Online." *Journal of Microbiology & Biology Education*, 23(1): e00236–21.
- Rand, Jenny R., and Sara C. Brushett.** 2023. "TikTok in the Virtual Health Promotion Classroom: Pivoting with Innovative Tools during the Pandemic." *SCHOLE: A Journal of Leisure Studies and Recreation Education*, 38(2): 92–93.
- Rejeb, Abderahman, Karim Rejeb, Andrea Appolloni, and Horst Treiblmaier.** 2023. "Foundations and knowledge clusters in TikTok (Douyin) research: evidence from bibliometric and topic modelling analyses." *Multimedia Tools and Applications*.
- Stantcheva, Stefanie.** 2022. "How to Run Surveys: A guide to creating your own identifying variation and revealing the invisible."
- Vaterlaus, J. Mitchell, and Madison Winter.** 2021. "TikTok: an exploratory study of young adults' uses and gratifications." *The Social Science Journal*, 1–20.
- Wang, Yunwen.** 2020. "Humor and camera view on mobile short-form video apps influence user experience and technology-adoption intent, an example of TikTok (DouYin)." *Computers in Human Behavior*, 110: 106373.
- Weekly, China Economic.** 2020. "How are impoverished counties determined? Where are the impoverished counties that have not yet been lifted out of poverty?"
- Yeung, Jessie, and Selina Wang.** 2023. "TikTok is owned by a Chinese company. So why doesn't it exist there? | CNN Business."